CLASS 10 MATHEMATICS • CHAPTER 4

Quadratic Equations

Exhaustive point-by-point presentation covering standard forms, factorisation, quadratic formula, and nature of roots directly from the textbook.

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Introduction & Historical Context

🏛 Prayer Hall Problem

- Carpet area = 300 m². Length is 1 m more than twice its breadth (x).
- Area equation: $x(2x + 1) = 300$.
- Leads to standard quadratic form: $2x^2 + x - 300 = 0$.

🏛 Historical Origins

- **Babylonians:** Solved equations of form $x^2 - px + q = 0$.
- **Brahmagupta (C.E. 598–665):** Gave explicit formula for $ax^2 + bx = c$.
- **Sridharacharya (C.E. 1025):** Derived the quadratic formula (completing the square).

Section 4.1 Live Check

Q1. How did we set up the algebraic equation for the prayer hall area in terms of its breadth x?

Q2. Which ancient Indian mathematicians contributed significantly to solving quadratic equations?

Q3. Why do quadratic equations frequently appear when modeling physical and geometric real-life problems?

Standard Form & Equation Checking

Standard Form Definition

An equation of the form $ax^2 + bx + c = 0$, where a, b, c are real numbers and $a \neq 0$, is called the standard form of a quadratic equation.

Degree 2 Polynomials

Any equation p(x) = 0 where p(x) is a polynomial of degree 2 represents a quadratic equation once simplified and arranged in descending order.

Simplification Caution

Always simplify equations fully (e.g., cubic terms cancelling out as in Example 2(iv)) before concluding whether an expression is quadratic.

Section 4.2 Live Check

Q1. What condition must be satisfied by coefficient 'a' for $ax^2 + bx + c = 0$ to be quadratic?

Q2. Can an equation initially appearing to have degree 3 (cubic) turn out to be quadratic upon expansion?

Q3. Why do we write quadratic equations in descending order of degrees?

Roots, Zeroes & Factorisation Method

🔑 Roots & Zeroes

- A real number α is a root of $ax^2 + bx + c = 0$ if $a\alpha^2 + b\alpha + c = 0$.
- Zeroes of quadratic polynomial = Roots of quadratic equation.
- A quadratic equation can have **at most two roots**.

✂ Splitting Middle Term

- Factorise $ax^2 + bx + c$ into two linear factors.
- Equate each linear factor to zero to find the roots.
- Example: $2x^2 - 5x + 3 = 0$ splits into $(2x - 3)(x - 1) = 0$.

Section 4.3 Live Check

Q1. What is the relation between the zeroes of a quadratic polynomial and the roots of a quadratic equation?

Q2. What is the maximum number of roots any quadratic equation can possess?

Q3. How do we find roots using middle-term splitting and linear factors?

The Discriminant & Quadratic Formula

Quadratic Formula & Discriminant

Roots are given by $x = (-b \pm \sqrt{(b^2 - 4ac)}) / (2a)$. The expression **$D = b^2 - 4ac$** is called the **discriminant**.

1. If $b^2 - 4ac > 0$: Two distinct real roots $(-b \pm \sqrt{(b^2 - 4ac)}) / 2a$
2. If $b^2 - 4ac = 0$: Two equal real roots $(-b / 2a, -b / 2a)$
3. If $b^2 - 4ac < 0$: No real roots

Section 4.4 Live Check

Q1. What is the discriminant of a quadratic equation $ax^2 + bx + c = 0$, and what letter represents it?

Q2. Under what condition on the discriminant does a quadratic equation have two equal real roots?

Q3. Why are there no real roots when the discriminant is negative?

Worked Exercise Highlights

Problem 1: Consecutive Positive Integers (Ex 4.1)

- Product of two consecutive positive integers is 306.
- Let integers be x and $x + 1$.
- $x(x + 1) = 306 \Rightarrow x^2 + x - 306 = 0$.
- Factorising: $(x + 18)(x - 17) = 0$.
- **Integers: 17 and 18.**

Problem 2: Equal Roots Parameter (Ex 4.3)

- Find k for $2x^2 + kx + 3 = 0$ having equal roots.
- For equal roots, discriminant $D = b^2 - 4ac = 0$.
- $k^2 - 4(2)(3) = 0 \Rightarrow k^2 - 24 = 0$.
- **$k = \pm\sqrt{(24)} = \pm 2\sqrt{(6)}$.**

Exercise Highlights Live Check

Q1. How do we represent consecutive positive integers algebraically?

Q2. What algebraic condition must hold for an equation to possess equal real roots?

Q3. Why do we get both positive and negative values when solving for parameters like k squared?

Chapter Recap & Checklist

Standard Form & Methods

- **Standard Form:** $ax^2 + bx + c = 0$ $(a \neq 0)$.
- **Factorisation:** Splitting middle term into linear factors.
- **Roots & Zeroes:** Roots of equation match polynomial zeroes.

Quadratic Formula

- **Formula:** $x = (-b \pm \sqrt{(b^2 - 4ac)}) / (2a)$.
- **Discriminant ($b^2 - 4ac$):** Determines whether roots are distinct real, equal real, or non-real.

THANK YOU

"The essence of mathematics is not to make simple things complicated, but to make complicated things simple." — S. Gudder

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IMAGE SOURCES

Attributions

Slide Visual Assets & Sources

- **Background Pattern:** Vibrant Educational Mathematics Pattern — View Source
- **Textbook Diagrams:** NCERT Class 10 Mathematics Textbook (Chapter 4 - Quadratic Equations).